

Instructor: Date:

Learner: Regional Branch:

Program: Center:

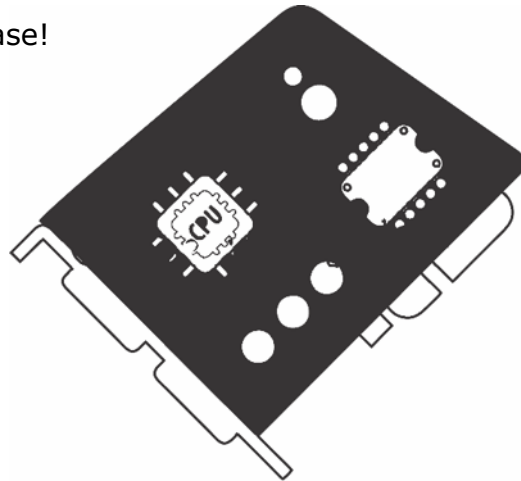
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Let's Start!

Inside a Computer's Case!



Learning Outcome:

Build the user interface, supported on the prototype assessment, determining the inputs and outputs required in the design and defining the navigation guidelines, according to the user's needs.

Identify different hardware components and peripheral devices.

Can Do Statements:

- I can explain what a bus is.
- I can talk about internal peripheral types.
- I can describe types of slots and types of cards as part of the internal peripheral devices.

Ask your partner:

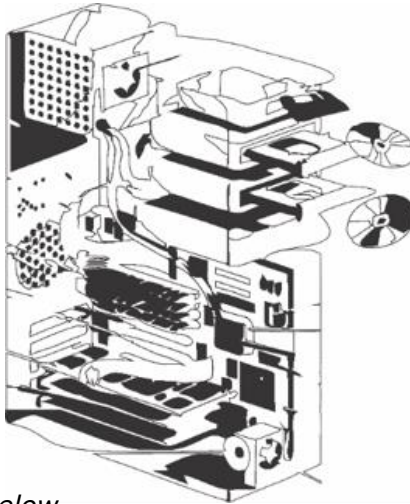
- What can we find inside a computer's CPU?
- What are the most common elements that you can include in a CPU to improve computer's performance?





Let's Explore!

Marianne needs to get a video card!



Instructions: Read the text below.

Marianne: I bought a new game, and I am looking forward to playing it, but my computer won't run it.

Bennett: You just probably need to add an expansion bus in your CPU.

Marianne: What is that?

Bennett: A bus is a subsystem that connects different peripheral devices using the same set of wires; they can be internal or external. Internal buses connect various components inside the CPU, and external buses are connected to components outside the unit. They transfer data which travels to the processor and the motherboard inside a computer. You should assemble an expansion bus to connect a video card to run your game.

Marianne: That sounds great! What else can I do to improve my computer?

Bennett: Well, you need to consider the internal peripheral types; they transfer information to the computer digitally instead of using analog signals. Internal peripherals always go inside the computer case. Slots and cards which give you extra benefits are also included in this category. Even though they're optional, they are not ruled out.

Marianne: Can you describe the types of slots and cards?

Bennett: Sure! The slots are designed to register the information that is provided by the card. The most popular types of slots are PCI, PCI Express, PCMCIA, and AGP. They all share similar functions, although they also have individual characteristics. On the other hand, we have different types of cards which include sound cards, network, cards and video cards. They offer certain features that allow you to do more things with your computer.

Marianne: To sum up, I need to get an expansion bus with a slot that matches up with a good video card. Then, my game will work as I expect.



**Let's Practice!**

Let's talk about internal peripheral types.

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: which transfer, which help, that go, who get, which give.

1. You need to take into account the internal peripheral types _____ you improve your computer's performance.
2. You must use Internal peripheral devices _____ information to the computer digitally instead of using analog signals.
3. The elements _____ inside the computer's case are the internal peripheral devices.
4. Slots and cards are elements _____ you extra benefits are included in this category, too.
5. Many people _____ these types of devices say that they are optional.

**Let's Practice!**

You need to know this information to play your video game.

Instructions: Read the paragraph. Underline all the sentences about definitions of bus.

An expansion bus is required in your CPU to play your video game. Don't you know what a bus is? This element is defined as a subsystem that connects different peripheral devices using the same set of wires. Don't ignore this information, they are divided into internal and external. Internal buses connect various components inside the CPU. On the other hand, external buses are connected to components outside the unit. In this case, data is transferred by these devices and they travel to the processor and the motherboard inside a computer.





Extension Activity

Let's find some more information about internal peripheral types.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about bus definition and internal peripheral types.
- Represent a bus in a graphic and explain its parts. Don't forget to mention the internal peripheral types in your graphic.
- Share your final product and explanation with your partners.

Self-Assessment

Yes

No

- I search information about bus definition and internal peripheral types.
- I sketch a representation of the bus.
- I explain in my graphic and the internal peripheral types to my partners.

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Target Words

- bus** (noun): subsystem that transfers data to the computer
e.g. Probably you just probable need to add an expansion bus in your CPU.
- analog** (adjective): when a digital device works with a continuous signal frequency
e.g. You need to take into account the internal peripheral types; they transfer information to the computer digitally instead of using analog signals.
- slot** (noun): a narrow space designed to insert something
e.g. The slots are designed to register the information that is provided by the card.
- processor** (noun): item that allows computers processing and functioning
e.g. They transfer data which travels to the processor and the motherboard inside a computer.
- card** (noun): a thin piece that contains digital information
e.g. The slots are designed to register the information that is provided by the card.
- assemble** (verb): to get items together for them to work
e.g. You should assemble an expansion bus to connect a video card to run your game.



Learning Points

Simple Present: Passive- Affirmative

You use the **simple present passive** to give more importance to the object and the action in a sentence rather than to the subject.

The **simple present passive** is used when the subject of the sentence can be omitted since it is implicit in the message. On the other hand, to mention the subject, you have to use **by** followed by it.

Let's compare these two sentences:

Active voice:

People use internal peripheral devices to improve a computer's performance.

Passive voice:

Internal peripheral devices are used (by people) to improve their computer's performance.

As you can see, in passive voice, the subject of the sentence "people" is not necessary because it is implicit.

This is the present simple passive structure:

Object of the sentence	verb 'to be'	main verb in past participle	complement
Peripheral devices	are	designed	to complement some features.

Examples

- Slots and cards which give you extra benefits are also included in this category.
- External buses are connected to components outside the unit.



Learning Points

Internal peripheral Types

You use **internal peripheral devices** to give extra functions to a computer by adding them inside the case.

Internal peripheral devices include different types of slots and cards that are connected to the computer thanks to a bus. They work in order to provide extra benefits to the computer. Let's take a look at some types of slots and cards:

Types of slots:

In this category we have Peripheral Component Interconnect (PCI), which is an advanced bus with the ability to include network cards, sound cards, and USB ports, among other features. In 2004, an upgraded version, PCI Express, was released. Another bus designed for this purpose was the PCMCIA, which stands for Personal Computer Memory Card International Association. Finally, we have the AGP that stands for Accelerated Graphics Port, used to attach graphic cards to a computer's motherboard.

Types of cards:

In this category we have video cards, sound cards, and network cards. Video cards are also known as graphic cards. Besides generating images, they offer some add-ons such as the ability to connect more monitors, TV tuner adapter or even 3D graphics rendering. Sound cards facilitate the input and output of audio for multimedia applications. Finally, network cards are used to establish a network connection among devices to communicate.

Types of slots	Types of cards
PCI / PCI Express	Video Card
PCMCIA	Sound Card
AGP	Network Card

Examples

- *The most popular types of slots are PCI, PCI Express, PCMCIA and AGP.*
- *They all share similar functions, although they also have individual characteristics.*
- *On the other hand, we have types of cards which include sound cards, network cards and video cards.*

